EVERY QUASI-COMPACT OPEN OF AN AFFINE SCHEME WITH ONE-DIMENSIONAL NOETHERIAN LOCAL RINGS IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites five cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Higher direct images along the inclusion of a quasi-compact open in the one-dimensional affine case vanish. Stacks Project, Varieties, Lemma lemma-affine (tag 09N9). Gives the vanishing of R^p j_*F which makes the Leray sequence collapse.
- Cohomology of quasi-coherent sheaves on affine schemes and the Leray spectral sequence. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero (tag 01XB); Cohomology, Lemma cohomology-lemma-apply-Leray (tag 01E3). Supply the vanishing on X and the spectral sequence transferring it to U.

Source reference. Original lemma 09N8: sources/stacks-chow/tags.txt, source lines 12206–12206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F4: sources/stacks-chow/tags.txt, source lines 1857–1857. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XF: sources/stacks-chow/tags.txt, source lines 2516–2516. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be an affine scheme all of whose local rings are Noetherian of dimension ≤ 1. Then any quasi-compact open $U \subset X$ is affine.*

Proof. Denote $j : U \to X$ the inclusion morphism. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_U$-module. By Lemma 09N8 the higher direct images $R^p j_* \mathcal{F}$ are zero. The $\mathcal{O}_X$-module $j_* \mathcal{F}$ is quasi-coherent (Schemes, Lemma 01LC). Hence it has vanishing

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

higher cohomology groups by Cohomology of Schemes, Lemma 01XB. By the Leray spectral sequence Cohomology, Lemma 01F4 we have $H^p(U, \mathcal{F}) = 0$ for all $p > 0$. Thus U is affine, for example by Cohomology of Schemes, Lemma 01XF. $\square$